

DBMS Full Syllabus (Complete)

1 Introduction to DBMS

What is DBMS?

Advantages and Disadvantages of DBMS

Applications of DBMS

DBMS vs File System

DBMS Architectures:

1-Tier, 2-Tier, 3-Tier Architecture

DBMS Users and Interfaces

Data Abstraction and Data Independence

2 Entity Relationship Model (ER Model)

Entity, Entity Sets, Attributes

Relationships & Relationship Sets

Primary Key, Composite Key, Candidate Key

Enhanced ER Model (EER)

Specialization, Generalization, Aggregation

Recursive Relationships

Minimization of ER Diagram

Mapping ER Model to Relational Model

Schema Integration

Practice: Draw ER diagrams for School, Hospital, Library

3 Relational Model

Concepts of Relations, Tuples, Attributes

Keys:

Candidate Key, Primary Key, Super Key, Alternate Key, Foreign Key

Functional Dependencies (FD)

Attribute Closure & Candidate Keys

Armstrong's Axioms for FD

Equivalence of FDs

Canonical Cover

Anomalies in Relational Model (Insertion, Update, Deletion)

Strategies for Schema Design

4 Normalization

Introduction to Normalization

Normal Forms:

1NF, 2NF, 3NF, BCNF

4NF, 5NF

Domain-Key Normal Form (DKNF)

Redundancy and Anomalies

Lossless Join Decomposition

Dependency Preserving Decomposition

Denormalization

Data Replication

Practice: Normalize relations for real-life databases (Students, Employees, Orders)

5 Relational Algebra & Relational Calculus

Basic Operators: SELECT, PROJECT, UNION, INTERSECTION, DIFFERENCE, CARTESIAN PRODUCT

Extended Operators: JOIN, DIVISION

Inner Join vs Outer Join

Join vs Nested Query

Tuple Relational Calculus (TRC)

Row-oriented vs Column-oriented data stores

Practice: Solve queries using relational algebra expressions

6 SQL & Queries

SQL Basics: CREATE, INSERT, UPDATE, DELETE, SELECT

Clauses: WHERE, GROUP BY, ORDER BY, HAVING

Joins: Inner, Left, Right, Full Outer

Subqueries, Nested Queries

Aggregate Functions: SUM, AVG, COUNT, MIN, MAX

Constraints: PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK, NOT NULL

Views, Indexes, Sequences

Practice: Build simple databases and write queries (School DB, Employee DB)

7 Transactions & Concurrency Control

Introduction to Transactions

ACID Properties (Atomicity, Consistency, Isolation, Durability)

Concurrency Control:

Lock-Based Protocols

Multiple Granularity Locking

Timestamp Ordering Protocols

Thomas Write Rule

Deadlock Prevention & Detection

Log-Based Recovery

Types of OLAP Systems

Practice: Simulate concurrent transactions and locks

8 File Organization & Indexing

File Organization Techniques

Heap, Sequential, Indexed Files

Indexing in Databases

Clustered vs Non-Clustered Indexes

B-Tree, B+ Tree

Bitmap Indexing, Inverted Index

Practice: Draw B-Tree / B+ Tree for sample keys

9 Advanced Topics

RAID Levels & Concepts

Distributed Database Systems

Query Optimization Techniques

RDBMS vs NoSQL (HBase, MongoDB)

Database Security Challenges

Mobile Database Management

Geographic Information Systems (GIS) & Future Trends

Data Warehouse & Data Mining:

Data Warehouse Architecture

Dimensional Data Modeling

Data Marts

Apache HBase & Hive:

Architecture and Working

Querying Hive

10 DBMS Interview & Miscellaneous Topics

SQL Interview Questions

PostgreSQL / PL/SQL Interview Questions

DBMS Set-based Practice Questions (Set1–Set10)

Impedance Mismatch

Database Objects (Tables, Views, Indexes, Triggers, Stored Procedures)

Suggested Practice Plan

Week Topics

- 1 Introduction, DBMS Concepts, ER Model**
 - 2 Relational Model, Keys, Functional Dependencies**
 - 3 Normalization (1NF–5NF, BCNF, DKNF)**
 - 4 Relational Algebra, Relational Calculus, SQL Basics**
 - 5 Advanced SQL (Joins, Subqueries, Views, Indexes)**
 - 6 Transactions, Concurrency Control, ACID Properties**
 - 7 File Organization, Indexing, B/B+ Trees**
 - 8 Advanced Topics (RAID, Distributed DB, Data Warehousing)**
 - 9+ Mini Projects / Practice Questions / Mock Interviews**
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Mini Project Ideas:

Student Management System

Hospital Database System

Library Management System

Online Shopping Database

Inventory Management

DBMS Practice Exercises & Mini Projects

1 Beginner Exercises (Concepts & ER Model)

Topics: DBMS Introduction, ER Model, Keys

Database Basics

List advantages and disadvantages of DBMS.

Identify types of DBMS: Relational, NoSQL, Hierarchical, Network.

ER Diagram Practice

Draw ER diagram for a School database:

Entities: Student, Teacher, Course

Relationships: Enroll, Teach

Primary & Foreign Keys

Draw ER diagram for Hospital database:

Entities: Patient, Doctor, Appointment, Ward

Enhanced ER Model

Implement specialization/generalization in ER diagrams.

Draw recursive relationships, aggregation, and minimization of ER diagram.

2 Relational Model Exercises

Topics: Relations, Keys, Functional Dependencies

Identify Candidate Key, Primary Key, Super Key, Alternate Key for a Student Table.

Given functional dependencies, find attribute closure.

Apply Armstrong's Axioms to derive all FDs.

Identify anomalies (Insertion, Update, Deletion) in sample tables.

3 Normalization Exercises

Topics: 1NF–5NF, BCNF, DKNF

Convert unnormalized table to 1NF.

Normalize table to 2NF and 3NF.

Identify BCNF violations in a sample database.

Apply Lossless Join & Dependency Preservation.

Practice Denormalization for query optimization.

Example: Employee Table → Normalize it to reduce redundancy

4 SQL Exercises

Topics: DDL, DML, Queries, Joins, Subqueries

Create a database SchoolDB and tables: Students, Courses, Enrollments.

Insert sample data into tables.

Queries:

Find all students enrolled in a specific course.

List teachers teaching more than 5 courses.

Update student marks for a course.

Delete a student who has graduated.

Advanced Queries:

Inner Join / Left Join / Right Join / Full Join examples

Nested queries to find highest marks per course

Aggregate functions: SUM, AVG, COUNT

5 Transactions & Concurrency Control

Topics: ACID, Locking, Deadlock

Simulate transaction scenario:

Two transactions updating the same account balance.

Show how locks prevent inconsistency.

Implement Timestamp Ordering Protocol manually on paper.

Identify ACID violations in sample transactions.

6 File Organization & Indexing

Topics: B-Tree, B+ Tree, Indexing

Draw B-Tree and B+ Tree for a set of keys.

Create clustered and non-clustered indexes in SQL.

Query using indexed vs non-indexed columns and observe performance.

7 Advanced DBMS Exercises

Topics: RAID, Distributed DB, Data Warehousing, Hive

Explain RAID levels and choose appropriate RAID for a given scenario.

Compare RDBMS vs NoSQL (HBase, MongoDB) with examples.

Design star schema for a Sales Data Warehouse.

Write HiveQL queries for sample dataset (Sales or Employee).

8 Mini Projects

Student Management System

ER diagram → Relational schema → SQL queries

Add / Update / Delete / Search student records

Library Management System

Track Books, Members, Borrow/Return transactions

Queries to find overdue books

Hospital Database

Patient, Doctor, Appointment tables

Reports: Patient visits, Doctor schedules

Online Shopping Database

Product, Customer, Orders, Payment

Queries: Top-selling products, Customer purchase history

Inventory Management

Track stock, suppliers, orders

Use normalization to avoid redundancy

 **Tips for Practice:**

Start with ER diagrams → Schema → SQL Queries

Normalize tables before writing SQL queries

Practice both theory questions + SQL coding

Use free databases like MySQL, PostgreSQL, SQLite for hands-on